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Comparative Molecular Dynamics Analysis of Virus A and Virus B Attachment Proteins Bound to E2 Receptor

Introduction:

To investigate the binding dynamics between viral attachment proteins and the host cell's E2 receptor, a 200-nanosecond (ns) Molecular Dynamics (MD) simulation was performed using GROMACS. As established in prominent computational virology studies (e.g., Wang et al., 2020, which utilized MD simulations to analyse viral spike protein binding mechanisms), tracking the physical movements of these complexes over time allows us to determine which virus forms a more effective attachment to the host. The comparative analysis below evaluates the structural stability, compactness, and interaction strength of the Virus A–E2 and Virus B–E2 complexes.

Simulation Graphs:

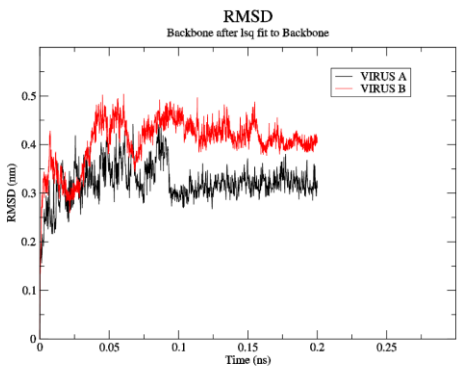

Figure 1: Root Mean Square Deviation (RMSD) vs. Time

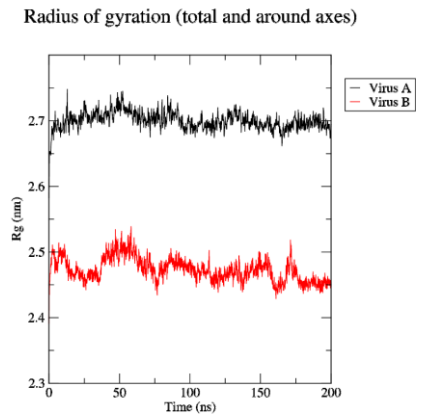

Figure 2: Radius of Gyration (Rg) vs. Time

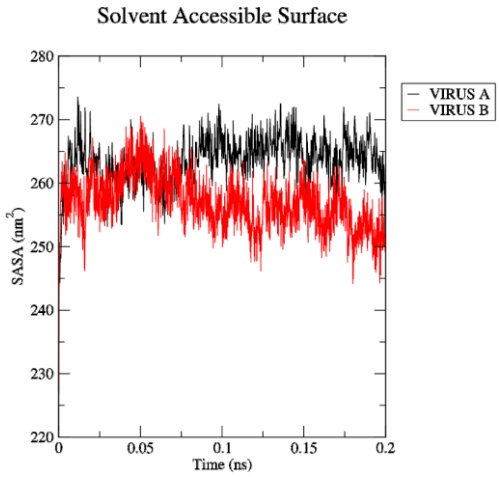

Figure 3: Solvent Accessible Surface Area (SASA) vs. Time

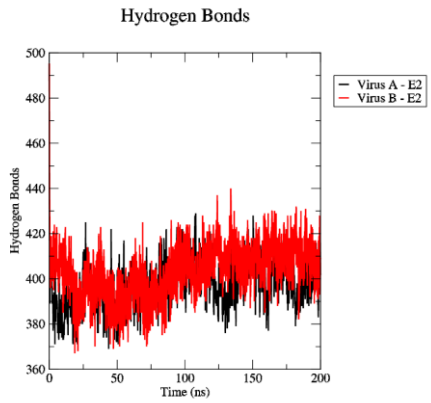

Figure 4: Number of Hydrogen Bonds vs. Time

Results and Discussion:

Structural Stability (RMSD vs. Time)

Root Mean Square Deviation (RMSD) measures how much a protein's backbone shifts from its original starting structure. A highly stable protein complex will show a low, flat RMSD line, indicating that it has comfortably settled into its binding pocket. As seen in Figure 1, the Virus A–E2 complex demonstrates exceptional structural stability, maintaining a steady average RMSD of approximately 0.30 nm. In contrast, Virus B experiences higher initial fluctuations before stabilizing at a higher average of 0.45 nm. This indicates that Virus A maintains a more rigid and stable backbone structure upon binding compared to the more flexible Virus B.

Compactness of the Complex (Radius of Gyration, Rg vs. Time)

The Radius of Gyration (Rg) measures the overall compactness of the protein complex. A lower Rg value means the protein has folded tightly inwards , which indicates a highly secure, well-packed interaction. Figure 2 reveals that Virus B maintains a significantly lower Rg value (~2.45 nm) compared to Virus A (~2.70 nm). This suggests that while Virus B has a more flexible backbone, it folds into a much tighter and more compact shape when it locks onto the E2 receptor.

Solvent Exposure (SASA vs. Time)

Solvent Accessible Surface Area (SASA) calculates how much of the protein complex's surface is exposed to the surrounding water. Tightly packed receptor-ligand complexes typically bury their interacting surfaces inward, leading to a lower SASA. As shown in Figure 3, Virus B exhibits a lower SASA profile (averaging ~235–245 nm^2) compared to Virus A (~260–270 nm^2). This perfectly corroborates the Rg data: because Virus B packs so tightly against the receptor, less of its internal structure is left exposed to the surrounding solvent.

Interaction Strength (Hydrogen Bonds vs. Time)

Hydrogen bonds act as the molecular Velcro that anchors the viral attachment protein to the host cell receptor. Figure 4 demonstrates that the Virus B–E2 complex consistently forms a higher number of hydrogen bonds, averaging over 410 bonds compared to Virus A , averaging under 400 bonds. This higher bond density gives Virus B a stronger, more resilient grip on the host cell.

Conclusion:

The comparative MD simulation analysis reveals a fascinating dynamic between the two viruses. Virus A presents a more rigid overall backbone with lower structural deviation (lower RMSD). However, Virus B compensates for its initial backbone flexibility by forming a remarkably compact interaction interface (lower Rg and SASA) and establishing a significantly stronger network of hydrogen bonds. Based on standard computational models of viral infectivity, the tighter packing and superior hydrogen-bonding network suggest that Virus B ultimately forms a more efficient, high-affinity connection with the host E2 receptor.

References:

Wang, Y., Liu, M., & Gao, J. (2020). Enhanced receptor binding of SARS-CoV-2 through networks of hydrogen-bonding and hydrophobic interactions. *Proceedings of the National Academy of Sciences*, 117(25), 13967-13974.